Probability and statistics

9A Genetics and evolution · Lesson 3 of 7

Date _____ Class _____

Today I can

- ☐ I can explain what probability is and write a probability as a percentage, a fraction and a decimal.
- ☐ I can calculate an experimental probability from results and compare it with a theoretical probability.

What happens today

1. Do Now: chance of rain
2. Learn: What statistics is for · Probability · and more
3. Practical: Investigation: two dice
4. Challenge, then the exit ticket

Student Book pages 10 to 11 · Workbook pages 7

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
statistics	collecting, analysing and presenting data
sample	a small part of a group that is measured to estimate the whole group
mean	the sum of the values divided by the number of values
median	the middle value when the values are in order
mode	the most common value in a set
probability	the chance of something happening
percentage	a number out of 100, shown with the % sign
fraction	a part of a whole, written as one number over another, e.g. 1/4
decimal	a number written with a decimal point, e.g. 0.25
experimental probability	a probability worked out from the results of an experiment
theoretical probability	a probability worked out by reasoning about all the possible outcomes

Quick Review

chance of rain

A weather app says there is a 76% chance of rain between 11:00 and 12:00 tomorrow.

1 What do you think 76% means here?

2 Write 76% as a fraction and as a decimal.

3 Is rain certain? How do you know?

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 Match each scientific term to its definition. Draw a line.

mean	the most common value in a set
median	the chance of something happening
mode	the sum of the values divided by the number of values
probability	the middle value in a set of values in order

2 Complete the probability scale. Fill in the missing values.

Percentage	Fraction	Decimal	Word
0%	0	0	impossible
25%	1/4		
50%		0.5	even chance
75%			likely
100%	1		certain

Core 2 of 3

Everyone does these. Answer on the card or under it.

3 Two dice investigation. Throw two dice, add the numbers, and tally the score. Do 50 throws.

a) Complete the table. Experimental probability = frequency divided by 50.

Score	Tally	Frequency	Fraction (out of 50)	Decimal	Percentage
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					

Core 3 of 3

Everyone does these. Answer on the card or under it.

3 *continued*

b) Which score was the mode (the most common) in your results?

Challenge 1 of 2

When Core is finished. Full sentences.

C1 Achondroplasia is an inherited condition. If neither parent has it, the theoretical probability of a child having it is 0.003%. If one parent has it, the probability is 50%. If both parents have it, the probability is 75%.

a) Write the probability for a child with one affected parent as a decimal and as a fraction.

b) A couple (one parent affected) has two children without the condition. Explain whether their third child is more likely to have it.

Challenge 2 of 2

When Core is finished. Full sentences.

C1 *continued*

c) Explain why these are theoretical probabilities and not experimental probabilities.

C2 Calculate the theoretical probability of throwing a number less than three with one dice. Give your answer as a fraction, a decimal and a percentage. Then explain why an experimental probability from 100 throws is more reliable than one from 10 throws.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is probability?

2 Write 5% as a decimal.

3 A bag has 20 red, 10 green, 5 blue and 15 yellow beads. Give the probability of taking a blue bead as a fraction and as a percentage.

Today I can: Goal 1 ☐ Goal 2 ☐